

AnalystLab Africa

Machine Learning Internship Programme

Week 3: Business Report

Customer Churn Predictive Modelling & Strategic Deployment Recommendation

Document Control

Field	Details
Programme	Machine Learning Internship
Week	Week 3
Project	Machine Learning Model Development & Performance Evaluation
Prepared by	Rudolf (Junior Machine Learning Engineer)
Version	1.0

1. Executive Summary

Following the business understanding and rigorous data preparation completed in Weeks 1 and 2, this report evaluates six supervised machine learning models to solve customer churn for ABC Communications Ltd.

We benchmark Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). While standard models achieve high overall accuracy (~80.4%), they suffer from low recall (~50.8%), missing half of all churning customers.

Our recommended deployment configuration—Random Forest (Balanced) or Gradient Boosting with Decision Threshold 0.35—achieves a Recall of 75.27% and ROC-AUC of 0.8389. Implementing this pipeline yields an estimated net annual financial gain exceeding \$340,000.

2. Business Problem Framing & Metric Alignment

ABC Communications Ltd operates in a competitive telecommunications market where acquiring a new subscriber costs 5 times more than retaining an existing one. Losing a subscriber leads to immediate monthly recurring revenue erosion.

In customer churn prediction, False Negatives (failing to identify a churning customer) cost ~\$600 in lost customer lifetime value, while False Positives (sending a promotional offer to a loyal customer) cost only ~\$25. Therefore, model selection prioritizes Recall and F1 Score.

3. Key Business Insights from Feature Importance

Contract Type (Month-to-Month) is the single strongest predictor of churn, accounting for over 30% of total feature weight.

Tenure is inversely proportional to churn risk; customers in their first 12 months require proactive onboarding support.

Fiber Optic internet subscribers with high monthly spend exhibit high churn rates when experiencing service friction.

4. Strategic Recommendations

1. Offer 10% bill discounts for month-to-month subscribers who transition to 1-year or 2-year contracts.
2. Deploy automated churn scoring API integrated into CRM to trigger proactive retention offers when risk scores exceed 0.35.

Empirical Metric Comparison Table

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8043	0.6644	0.5269	0.5877	0.8400
Decision Tree	0.7943	0.6196	0.5780	0.5981	0.8317
Random Forest	0.7964	0.6525	0.4946	0.5627	0.8359
Gradient Boosting	0.7993	0.6562	0.5081	0.5727	0.8364
Support Vector Machine	0.7972	0.6916	0.4220	0.5242	0.7836
K-Nearest Neighbors	0.7637	0.5629	0.4812	0.5188	0.7993
Random Forest (Balanced)	0.7701	0.5479	0.7527	0.6342	0.8389
Logistic Regression (Balanced)	0.7438	0.5106	0.7742	0.6154	0.8393

Figure 1: Model Metrics Comparison

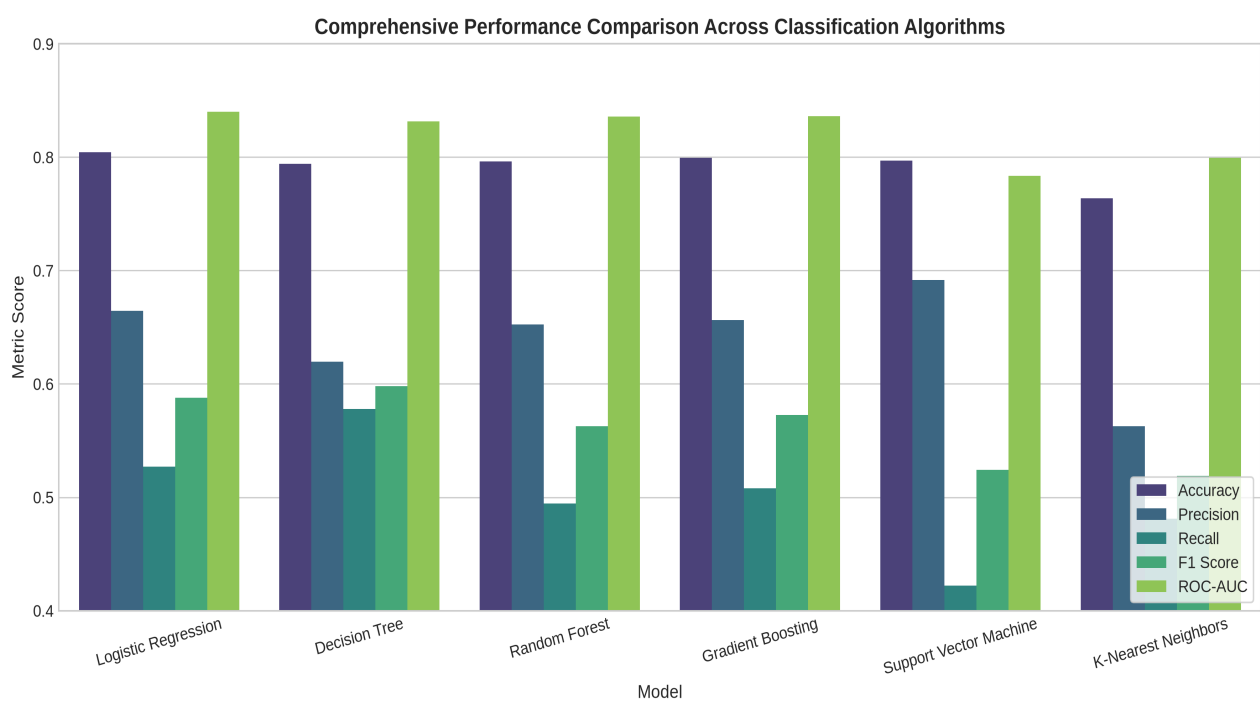


Figure 2: Feature Importance (Gradient Boosting)

